Final Seminar Review

Image Steganography Project

Team:
GOUTAM R DUBASHI 1MJ20CS072

Project Coordinator :
Mrs. Sophia G
Assistant Professor , MVJCE

Semester – VI
Academic Year – 2022-2023 (EVEN)

Content

COLLEGE OF ENGINEERING
Established In 1982

Abstract

Image steganography is a technique that enables the covert transmission of information by embedding it within digital images. In today's interconnected world, where data privacy and security are paramount, steganography serves as an effective tool to ensure confidentiality by hiding sensitive data in plain sight. This abstract provides an overview of image steganography, its underlying principles, and its significance in the field of information security.

Introduction

- Steganography involves the seamless integration of secret data, known as the payload, into the pixels of an image without perceptible alterations to the visual appearance of the cover image. Various methods have been developed to achieve this objective, ranging from the least significant bit (**LSB**) substitution to more advanced techniques like discrete cosine transform (**DCT**) and spread spectrum modulation. These methods exploit the characteristics of digital images, such as color channels, pixel intensities, and frequency domains, to embed the payload in a manner that remains undetectable to the human eye.
- The key advantages of image steganography lie in its imperceptibility and its resistance to detection. As the payload remains hidden within the image, unauthorized individuals are unlikely to even suspect the presence of concealed information. Additionally, steganographic techniques can withstand scrutiny by detection algorithms that aim to identify hidden data. This attribute makes steganography a valuable tool in various domains, including military communications, covert intelligence operations, digital watermarking, and copyright protection.

Problem Statement

- ***Capacity versus Security:*** One of the fundamental challenges in image steganography is finding the right balance between the amount of data (payload) that can be concealed within an image and the level of security provided. Increasing the payload capacity often leads to visible alterations in the cover image, making it more susceptible to detection.
- ***Robustness against Attacks:*** Image steganography techniques should be designed to withstand various attacks aimed at uncovering the hidden information. Steganalysis, the process of detecting steganographic content, has seen advancements, necessitating the development of more robust steganographic methods.
- ***Embedding in Different Image Formats:*** Images come in various formats, such as JPEG, PNG, and BMP, each with its own characteristics and compression techniques. Designing steganographic algorithms that are compatible with different image formats.
- ***Real-Time Steganography:*** With the increasing demand for real-time communication and multimedia applications, the need for efficient and fast steganographic techniques becomes crucial.
- ***Steganography and Deep Learning:*** As deep learning techniques continue to advance, exploring their application in image steganography becomes essential.

Advantages

1.Covert Communication: Image steganography allows for the secret transmission of information in a covert manner. By embedding data within digital images, steganography provides a way to communicate without arousing suspicion or drawing attention to the existence of hidden information.

2.Security through Obscurity: Image steganography relies on the concept of security through obscurity. The hidden information remains concealed within the image, making it difficult for unauthorized individuals to even suspect its presence

3.Resistance to Detection: Steganographic techniques can be designed to withstand various detection methods and attacks. By carefully embedding the payload within the image, steganography can evade visual inspection and automated detection algorithms.

4.Inconspicuous Transmission: Images are a common and widely accepted form of media in today's digital world. By using image steganography, sensitive information can be transmitted through normal channels, such as email attachments or image sharing platforms, without raising suspicion

5.Capacity for Large Payloads: Digital images can provide significant capacity for hiding data. The pixels in an image offer numerous possibilities for embedding information, allowing for the concealment of large payloads.

Disadvantages

- ***Limited Capacity***: While image steganography provides the capability to hide data within images, there is a limit to the amount of information that can be concealed without significantly degrading the image quality or raising suspicion.
- ***Susceptibility to Attacks:*** While steganography aims to remain undetectable, it is not immune to attacks. Steganalysis techniques have evolved to detect steganographic content, and adversaries can employ statistical, visual, or machine learning-based methods to uncover hidden information.
- ***Fragility of Hidden Information:*** Concealed data within images can be vulnerable to unintentional or intentional image modifications. Common image editing operations, such as compression, resizing, or filtering, may inadvertently alter or remove the hidden information, rendering it unrecoverable.
- ***Dependency on Image Formats:*** Different image formats have distinct characteristics and compression techniques, which can affect the effectiveness and security of steganographic methods. Techniques that work well with one image format may not be as efficient or secure with another.
- ***Computational Overhead:*** Embedding and extracting hidden information within images can utationally intensive, especially when dealing with large image files or complex raphic techniques. The process of encoding and decoding the hidden data may gnificant computational resources,

Applications

- ***Secure Communication:*** One of the primary applications of image steganography is secure communication. By concealing sensitive information within digital images, steganography provides a covert means of transmitting confidential data without raising suspicion.
- ***Data Privacy and Confidentiality:*** Image steganography can be used to protect data privacy and maintain confidentiality. Sensitive information, such as personal identification details, financial records, or medical records, can be hidden within images, ensuring that only authorized recipients can extract and access the concealed data.
- ***Digital Watermarking:*** Image steganography techniques can be utilized for digital watermarking, where information is embedded within images to establish ownership or verify authenticity. Watermarks can be used to protect copyrights, prevent unauthorized distribution, or trace the source of leaked or pirated content.
- ***Covert Intelligence Operations:*** Steganography plays a crucial role in covert intelligence operations, where secret information needs to be transmitted or stored securely. By hiding data within images.
- ***Digital Forensics:*** Steganography plays a role in digital forensics, where investigators analyze digital media to uncover hidden information. In cases involving digital evidence, steganalysis [illegible]s are employed to detect and extract concealed data from images.

Literature Survey

Paper 1

1.Paper Title: ***"A Survey on Image Steganography Techniques and Evaluation Methods"*** **Authors:** V. K. Patil, N. V. Patil, S. S. Sutar Summary: This survey paper provides an extensive overview of various image steganography techniques, including LSB substitution, spatial domain techniques, transform domain techniques, and adaptive methods. It also discusses different evaluation metrics used to assess the performance and security of steganographic algorithms.

2.Paper Title: ***"A Comprehensive Survey of Image Steganography Techniques" (2020)*** **Authors:** R. S. Bhosale, S. B. Ghotkar Summary: This survey paper presents a comprehensive analysis of image steganography techniques, including both spatial and transform domain methods. It explores different embedding and extraction techniques, along with their advantages, limitations, and security implications. The paper also discusses challenges and open research areas in image steganography

Paper 2

3.Paper Title: ***"Recent Advances in Image Steganography: A Survey" (2018)*** **Authors:** V. Thakur, D. Kapoor, A. K. Sharma Summary: This survey paper focuses on recent advancements in image steganography techniques. It covers topics such as evolutionary algorithms, neural networks, and deep learning-based approaches for steganography. The paper provides an in-depth analysis of each technique, their strengths, weaknesses, and potential applications.

4.Paper Title: "***A Survey of Steganographic Techniques in Spatial and Transform Domain"*** **Authors:** P. Chandramouli, D. S. Jayalakshmi Summary: This survey paper presents an overview of steganographic techniques in both spatial and transform domains. It discusses classic methods such as LSB substitution, as well as more advanced techniques like wavelet and DCT-based methods. The paper compares the performance, robustness, and security of these techniques and highlights their applications.

5.**Paper Title:** ***"A Survey on Deep Learning Techniques for Steganography and Steganalysis"*** (2021) **Authors**: A. Singh, S. Sood, R. Gupta Summary: This survey paper focuses on the integration of deep learning techniques in image steganography and steganalysis. It reviews recent studies that leverage deep neural networks for embedding and detecting hidden information. The paper discusses various architectures, training methodologies, and feature extraction techniques used in deep learning-based steganography and steganalysis.

6.**Paper Title:** ***"A Survey on Secure Image Steganography Techniques"*** (2020) **Authors:** P. Prashanthi, S. Vishwanath Summary: This survey paper provides an overview of secure image steganography techniques, emphasizing the importance of encryption in steganographic methods. It explores various encryption-based steganography techniques and discusses their effectiveness in terms of security and robustness. The paper also addresses the challenges and future research directions in secure image steganography.

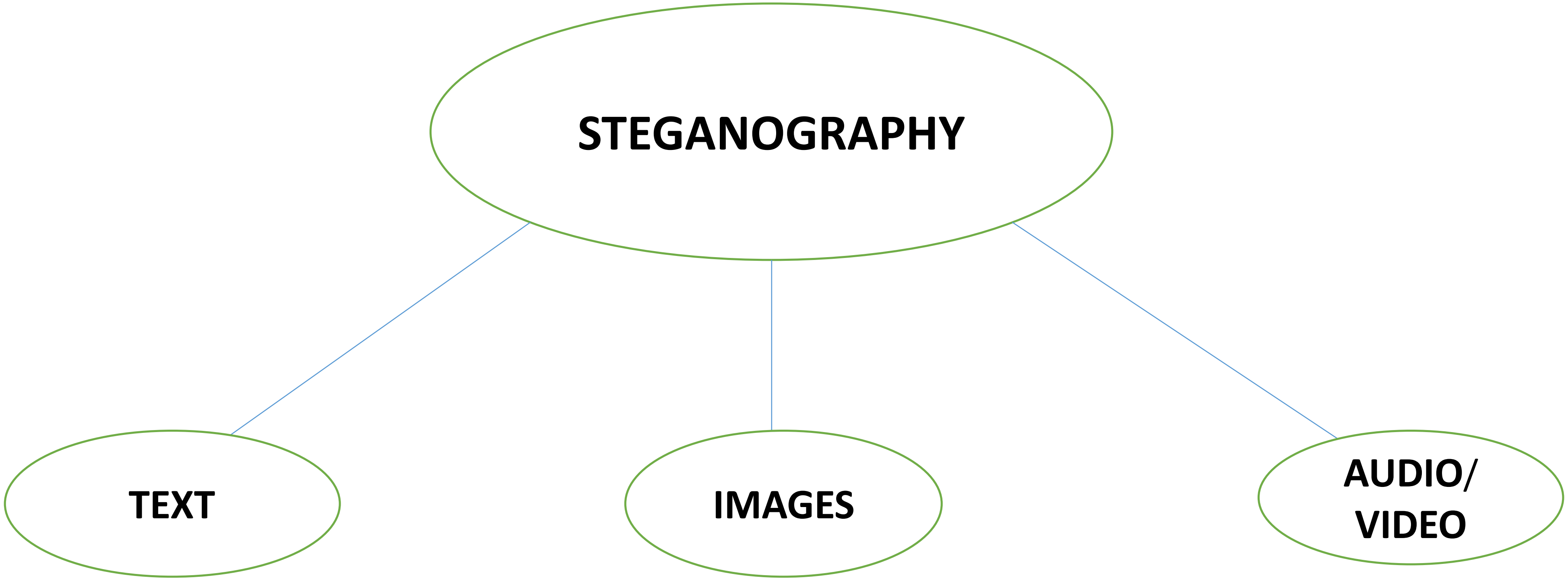
SYSTEM SPECIFICATION

H/W SPECIFICATION

1. Processor : Any Processor above 500 MHz
2. Ram : 512Mb.
3. Hard Disk : 10 GB.
4. Compact Disk : 650 Mb.
5. Input device : Standard Keyboard and Mouse.
6. Output device : VGA and High Resolution Monitor
7. LAN connected Systems

S/W SPECIFICATION

Operating System	:Windows Family.
Language	: JAVA
Tools	: Apache-Net-Beans ,Java JDK


STEGANOGRAPHY
TEXT
IMAGES
AUDIO/
VIDEO

Architecture

What Images are made up of?:-

Images are made up of lots of little dots called **pixels**. Each pixel is represented as **3** bytes - one for **Red**, one for **Green** and one for **Blue**.

248	201	3
11111000	11001001	00000011

Each byte is interpreted as an integer number, which is how much of that color is used to make the final color of the pixel.

248	201	3	OrangeColor

Image domain:

- The difference between two colors that differ by one bit in either one red, green or blue value is impossible detect for a human eye.
- So we can change the least significant (last) bit in a byte, we either add or subtract one or more values from the value it represents.
- This means we can overwrite the last bit in a byte without affecting the colors it appears to be.

A common approach of hiding data within an image file is **Least Significant Bit (LSB)** Substitution.

Suppose we have the following binary representation for the Cover Image.

10010101 00001101
10010110 00001111

Suppose we want to "hide" the following 4 bits of data: 1011, we get the following,

1001010**1** 0000110**1**
1001011**0** 0000111**1**

Where the each data bits are accommodated in the least significant bits of each byte of the image

- Open

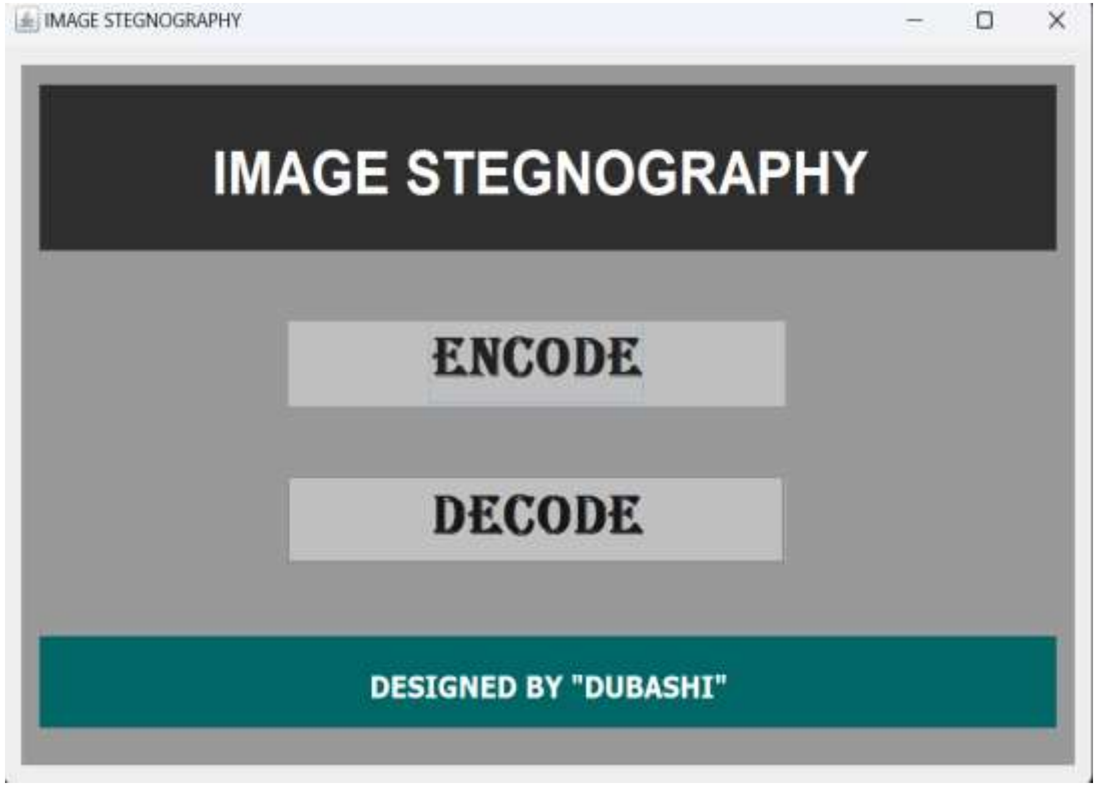
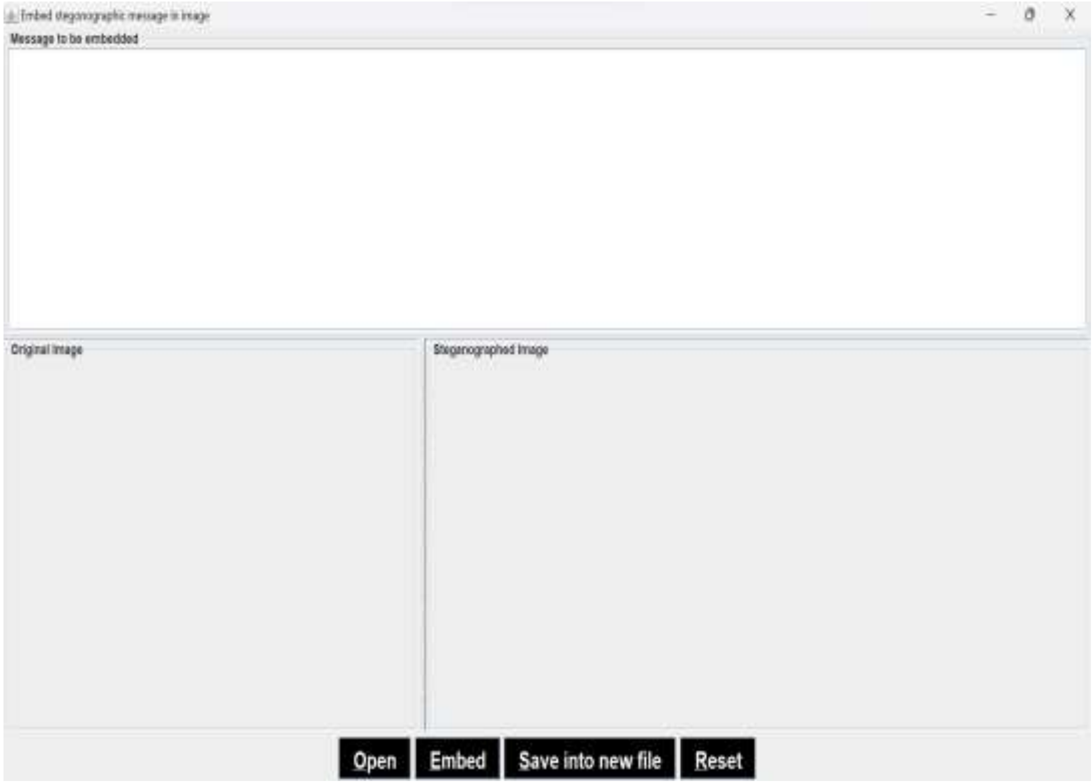
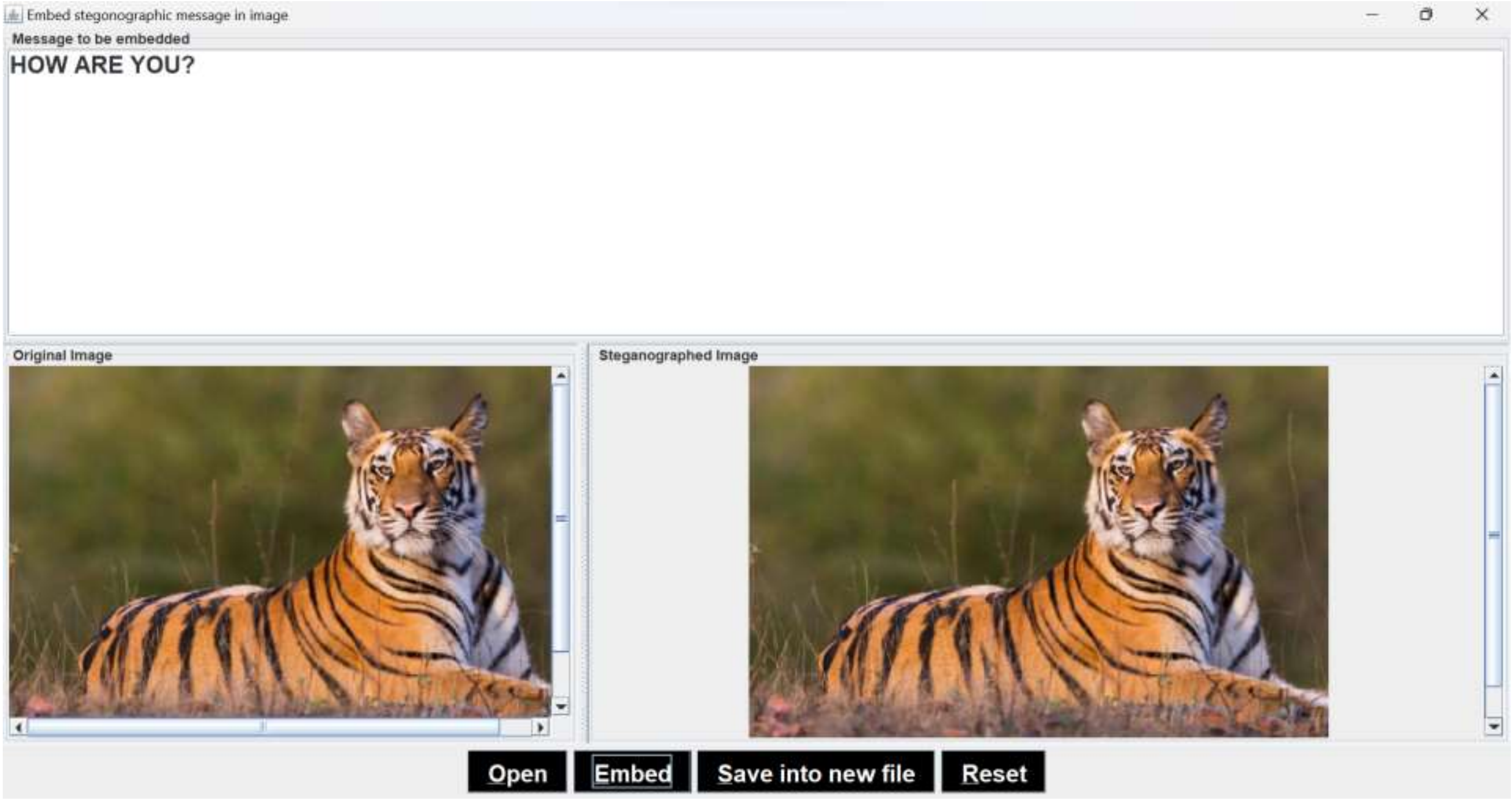


IMAGE STEGNOGRAPHY
IMAGE STEGNOGRAPHY
ENCODE
DECODE
DESIGNED BY "DUBASHI"


Message to be embedded
Original Image
Steganographed Image
Open
Embed
Save into new file
Reset

- Encode


Embed stegonographic message in image
Message to be embedded
HOW ARE YOU?
Original Image
Steganographed Image
Open
Embed
Save into new file
Reset

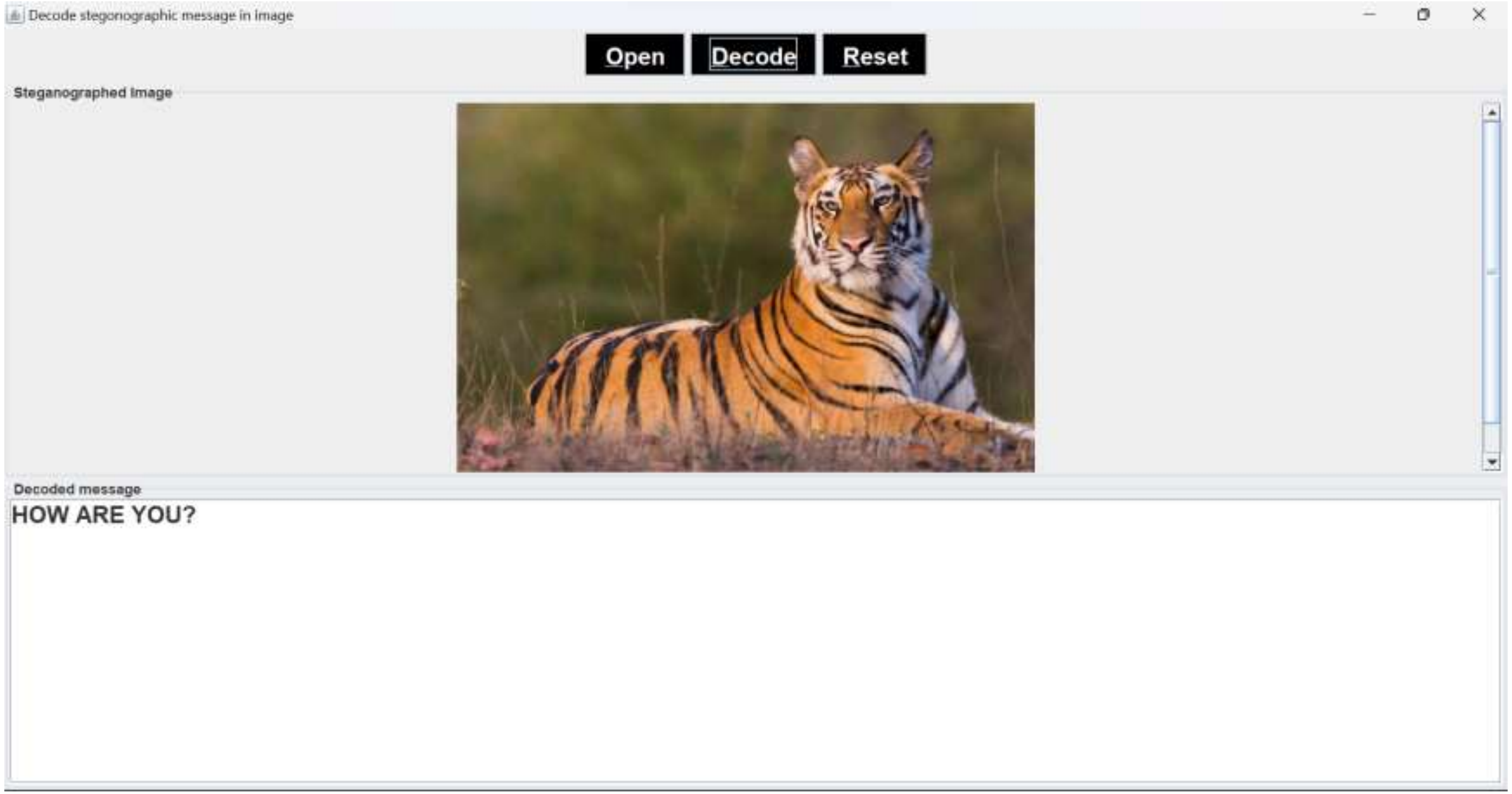
DIFFERENT METHODS:

- PIXEL INDICATOR
- SCC(STEGO COLOR CYCLE)
- Triple-A
- MAX-BIT

- **<u>INVERTED PATTERN</u>**

This inverted pattern (IP) LSB substitution approach uses the idea of processing secret messages prior to embedding.
In this method each section of secret images is determined to be inverted or not inverted before it is embedded.
In addition, the bits which are used to record the transformation are treated as secret keys or extra data to be re- embedded.

- Decode


Decode stegonographic message in image
Open
Decode
Reset
Steganographed Image
Decoded message
HOW ARE YOU?

Summary

- In conclusion, image steganography offers a covert and secure means of transmitting information by embedding it within digital images. Its ability to maintain visual fidelity while concealing sensitive data makes it an important tool in the realm of information security. As technology advances, researchers and practitioners must continuously develop robust steganographic algorithms and detection methods to stay ahead in this ever-evolving cat-and-mouse game between concealment and detection.

THANK YOU